

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

Mid-Semester Test
EC-2 Makeup

Course No. : AIMLCZC416
Course Title : Mathematical Foundations for Machine Learning
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 04-01-2026

No. of Pages = 2

No os questions = 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Notation used here are as per the text book.

Q1.(a) Find the Echelon form of the matrix $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ and hence find the determinant of

A. [2M]

(b) Find the inverse of the matrix A, and using this inverse solve the system of linear equations

$$AX = b \text{ where } b = \begin{bmatrix} -2 \\ 2 \\ 1 \\ -1 \end{bmatrix}. \text{ [2M]}$$

Q2. Consider the matrix $A = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 2 & -2 \\ 3 & -3 & 4 \end{bmatrix}$.

- (a) Show that the set $V = \{X \in R^3 : AX = 0\}$. Prove that V is a subspace of R^3 . [2M]
- (b) Find a basis and dimension of V . [2M]
- (c) Is it possible to find a one-dimensional subspace of V . If so find one such a subspace U , then find a basis of U . [2M]

Q3. Find the singular value decomposition of the matrix $A = \begin{bmatrix} 5 & 0 \\ 5 & 0 \\ 0 & -10 \end{bmatrix}$. [6M]

Q4. Let $A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$.

- (a) Prove that A is a symmetric, positive definite matrix. [2M]
- (b) Using this matrix A, define an inner product on R^4 . Using this inner product, compute the

distance between $\begin{bmatrix} -2 \\ 0 \\ 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix}$. [2M]

Q5. Let $f(x, y, z) = xy^2 + yz^2 + zx^2$

- (a) Find the gradient of $f(x, y, z)$. [2M]
- (b) Find all the critical points of the function $g(x, y) = f(x, y, 0)$ [1M]
- (c) Is it possible to use the second derivative test here? Why? [2M]

Q6. (a) Find the Taylor polynomial of degree 2 of the function $f(x) = \exp(\sin x^2)$ at 0. [4M]

- (b) Just by observing the function $f(x)$ defined in (a), write down the smallest and largest values of $f(x)$. [1M]
